

Total No. of Questions : 4]

SEAT No. :

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PF1

Apr-26/FE/Insem-1

F.E. (Insem)

ENGINEERING MATHEMATICS - II

(2019 Pattern) (Credit System) (Semester - II) (107008)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Attempt Q.1 or Q.2 and Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Use of logarithmic tables, slide rule, mollier chart, electronic pocket calculator and steam tables is allowed.
- 5) Assume suitable data, if necessary.

Q1) a) Solve $(x + y - 2)dx + (x - y + 4)dy = 0$. [5]

b) Solve $(x^2 + y^2 + x)dx + xydy = 0$. [5]

c) Solve $(1 + y^2)\frac{dx}{dy} + x = e^{-\tan^{-1}(y)}$. [5]

OR

Q2) a) Solve $\frac{dy}{dx} - xy = -y^3 e^{-x^2}$. [5]

b) Solve $\sec^2(y)\frac{dy}{dx} + 2x \tan(y) = x^3$. [5]

c) Solve $(x^2 y - 2xy^2)dx + (-x^3 + 3x^2 y)dy = 0$. [5]

P.T.O.

Q3) a) A body at temperature 80°C is placed in a room whose temperature is 20°C and cools to 40°C in 10 minutes. Find temperature of the body after 15 minutes from beginning. [5]

b) The charge 'Q' on the plate of a condenser of capacity 'C' charged through a resistor 'R' by a steady voltage 'V'. If $Q = 0$ at time $t = 0$ show that $Q = CV [1 - e^{-t/Rc}]$. Find current flowing into the plate. [5]

c) A body of mass m falling from rest is subjected to the force of gravity and an air resistance proportional to the square of the velocity (kV^2). If it falls through a distance x and possesses velocity v at that instant. Prove that

$$\frac{2kx}{m} = \log \left(\frac{a^2}{a^2 - v^2} \right) \text{ where, } (mg = ka^2). \quad [5]$$

OR

Q4) a) Find orthogonal trajectories of the family of curves given by $x^2 - y^2 = c^2$. [5]

b) An electrical circuit consists of a battery E , a resistance R and an inductance L . Find I as a function of t . Also find how long will be, before the current (I) reached one half of its maximum value, if $E = 6$ volts, $R = 100$ ohms & $L = 0.1$ henry? [5]

c) A long hollow pipe has an inner diameter of 10cm and outer diameter of 20cm. The inner surface is kept at 200°C and the outer surface at 50°C . The thermal conductivity $k = 0.12$. Find the amount of heat lost per second from a centimeter length pipe. Also find heat lost per minute from 10 meter length pipe. [5]

x x x